

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM TASK (5 Questions)

Course Code:	DSA0607	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	Assessment:	Assessment Tool 1 – Industry Problem Task
Weightage:	40% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	<i>Apply appropriate visualization methods to represent distributions, proportions, and relationships in data.(BL3)</i>		
Student Name:	B Akshaya	Reg. No.:	192424336
Section / Batch:	2024/D	Date:	25.07.2026

Code of Conduct

I B Akshaya(192424336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B Akshaya

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Categorical Data Visualization	Comparing category-wise revenue, proportions, and seasonal trends using bar, grouped, and stacked charts	1	10
Distribution Analysis	Understanding defect distribution and variability using histograms, boxplots, and violin plots	2	10
Relationship Visualization	Exploring relationships between variables using scatter plots, heatmaps, and multivariate techniques	3	10
Geospatial & Temporal Visualization	Analyzing delivery performance using maps, time-series plots, and heatmaps	4	10
Distribution Comparison	Comparing patient data distributions using ridgeline plots, density plots, and boxplots	5	10
GRAND TOTAL:			50 Marks

Annexure A – Industry Problem Task

1.A university wants to analyze the academic performance of students across different departments (Computer Science, Mechanical, Civil, Electronics, etc.). The administration needs to compare average scores, identify top-performing departments, and examine variations in student results.

Task: Design suitable visualizations to represent student scores and performance distributions. Justify the use of bar plots, grouped bars, histograms, and boxplots. Explain how these visualizations help identify performance trends and support academic improvement strategies.

Aim:

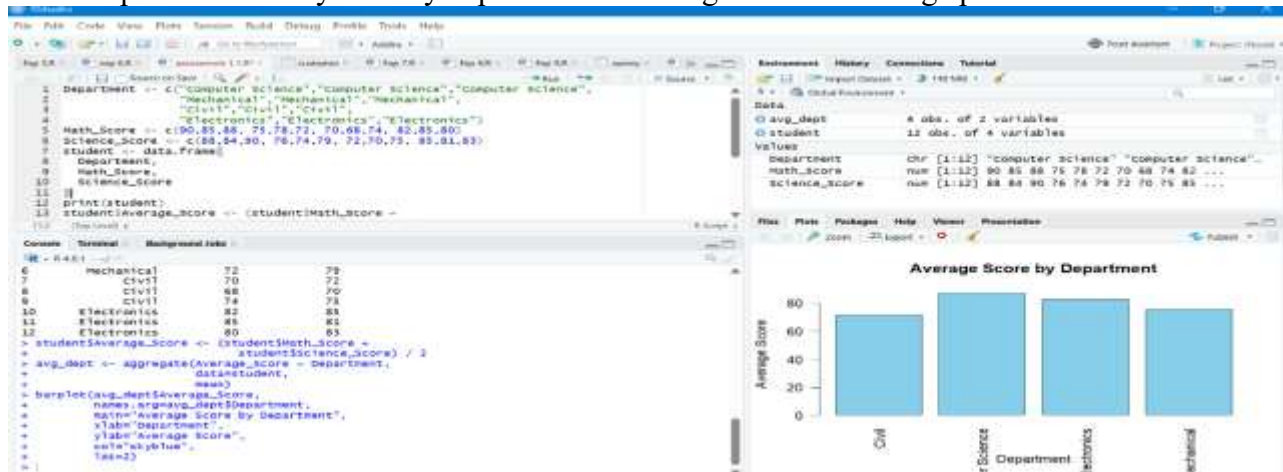
The university can use bar plots, grouped bar charts, histograms, and boxplots to compare student performance across departments.

1. Bar Plot

A bar plot can display the average score of each department.

- X-axis → Department (Computer Science, Mechanical, Civil, Electronics)
- Y-axis → Average Score
- Each bar represents one department.
- The tallest bar identifies the top-performing department.
- It makes comparison of departmental performance easy.

Use: Helps the university identify departments with high and low average performance.



2. Grouped Bar Plot

A grouped bar plot can compare scores in different subjects across departments.

Example:

- X-axis → Department
- Y-axis → Average Score
- Groups → Mathematics, Programming, Science, etc.

Use: Helps identify which departments perform better in particular subjects and where additional academic support is required.

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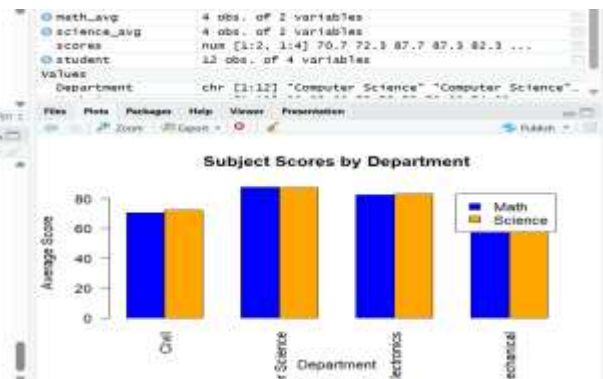
5 math_score <- c(80,85,88, 75,78,72, 70,68,74, 82,83,80)
6 science_score <- c(85,84,90, 76,74,79, 72,70,75, 85,81,83)
7 student <- data.frame(
8   Department,
9   Math_Score,
10  Science_Score
11 )
12 print(student)
13 student$Average_Score <- (student$Math_Score +
14   student$Science_Score)/2

```

```

# R 4.0.3
#> data=student,
#> mean()
#> science_avg <- aggregate(Science_Score ~ Department,
#>   data=student,
#>   mean)
#> scores <- rbind(math_avg$Math_Score,
#>   science_avg$Science_Score)
#> barplot(scores,
#>   beside=TRUE,
#>   names.arg=math_avg$Department,
#>   main="Subject Scores by Department",
#>   xlab="Department",
#>   ylab="Average Score",
#>   col=c("blue","orange"),
#>   legend.text=c("Math", "Science"),
#>   las=2)

```



3. Histogram

A histogram represents the distribution of student scores.

- X-axis → Score ranges such as 50–60, 60–70, 70–80, etc.
- Y-axis → Number of students

Use: Shows whether most students have high, average, or low scores. It also helps identify the overall pattern and spread of academic performance.

```

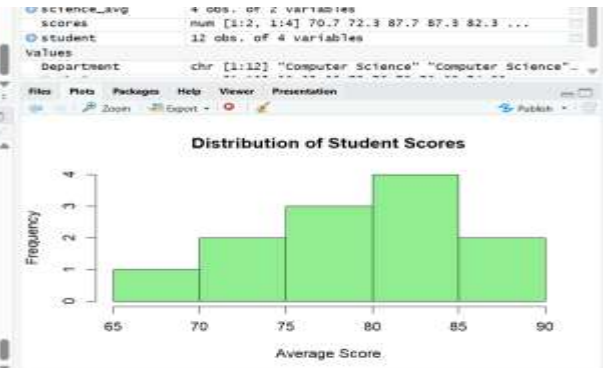
hist(student$Average_Score,
  main="Distribution of Student Scores",
  xlab="Average Score",
  ylab="Frequency",
  col="lightgreen",
  border="black")

```

```

# R 4.0.3
#> res <- rbind(math_avg$Math_Score,
#>   science_avg$Science_Score)
#> plot(scores,
#>   beside=TRUE,
#>   names.arg=math_avg$Department,
#>   main="Subject Scores by Department",
#>   xlab="Department",
#>   ylab="Average Score",
#>   col=c("blue","orange"),
#>   legend.text=c("Math", "Science"),
#>   las=2)
#> t(student$Average_Score,
#>   main="Distribution of Student Scores",
#>   xlab="Average Score",
#>   ylab="Frequency",
#>   col="lightgreen",
#>   border="black")

```



4. Boxplot

A boxplot can compare the distribution of student scores across departments.

- X-axis → Department
- Y-axis → Student Score
- Displays median, quartiles, spread, and outliers.

Use: Helps identify departments with consistent performance, large variations in marks, and unusually high or low student scores.

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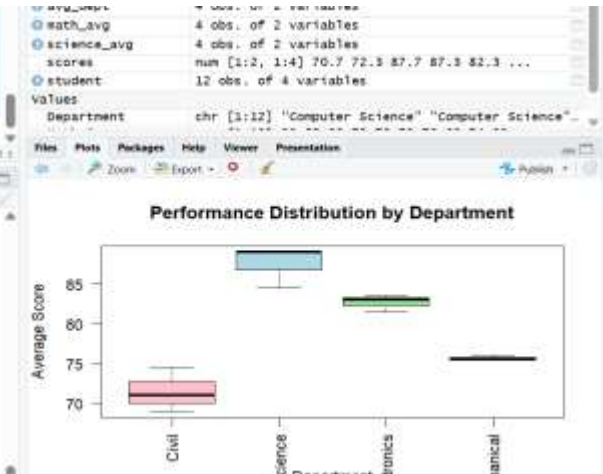
#> boxplot(Average_Score ~ Department,
#>   data=student,
#>   main="Performance Distribution by Department",
#>   xlab="Department",
#>   ylab="Average Score",
#>   col=c("pink","lightblue",
#>     "lightgreen","orange"),
#>   las=2)

```

```

# R 4.0.3
#> hist(student$Average_Score,
#>   main="Distribution of Student Scores",
#>   xlab="Average Score",
#>   ylab="Frequency",
#>   col="lightgreen",
#>   border="black")
#> boxplot(Average_Score ~ Department,
#>   data=student,
#>   main="Performance Distribution by Department",
#>   xlab="Department",
#>   ylab="Average Score",
#>   col=c("pink","lightblue",
#>     "lightgreen","orange"),
#>   las=2)

```



Conclusion

These visualizations help the university compare departmental performance, identify top-performing departments, detect variations and outliers, and understand overall score distributions. The findings can support academic improvement strategies such as additional training, tutoring, curriculum changes, and targeted support for lower-performing departments.

2.A supermarket chain collects data on customer purchases across various product categories and seasons. The management wants to understand spending patterns, compare sales distributions, and identify categories with the highest contribution to revenue.

Task: Recommend appropriate visualization techniques such as stacked bars, heatmaps, density plots, and violin plots. Explain how these visualizations help reveal customer preferences, seasonal trends, and business opportunities.

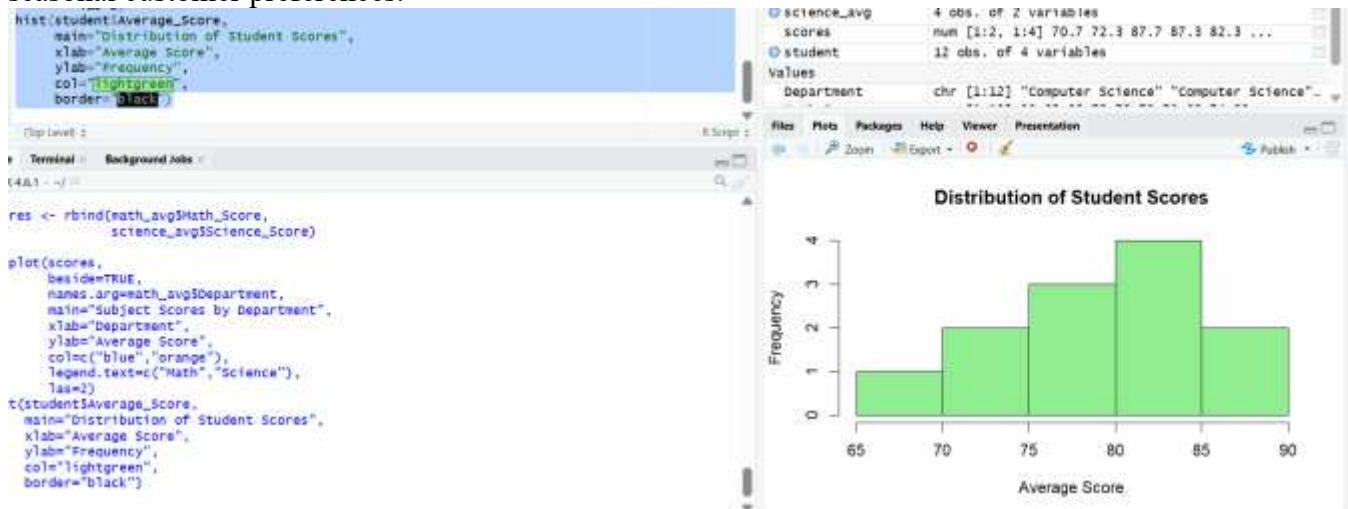
Aim: A supermarket can use stacked bar charts, heatmaps, density plots, and violin plots to understand customer spending patterns across different product categories and seasons.

1. Stacked Bar Chart

A stacked bar chart can display sales of different product categories across seasons.

- X-axis → Season (Summer, Winter, Spring, Autumn)
- Y-axis → Sales
- Stacks → Electronics, Clothing, Groceries, Home
- Each stack represents the contribution of a product category.

Use: Helps identify which product categories contribute the most revenue during each season and reveals seasonal customer preferences.

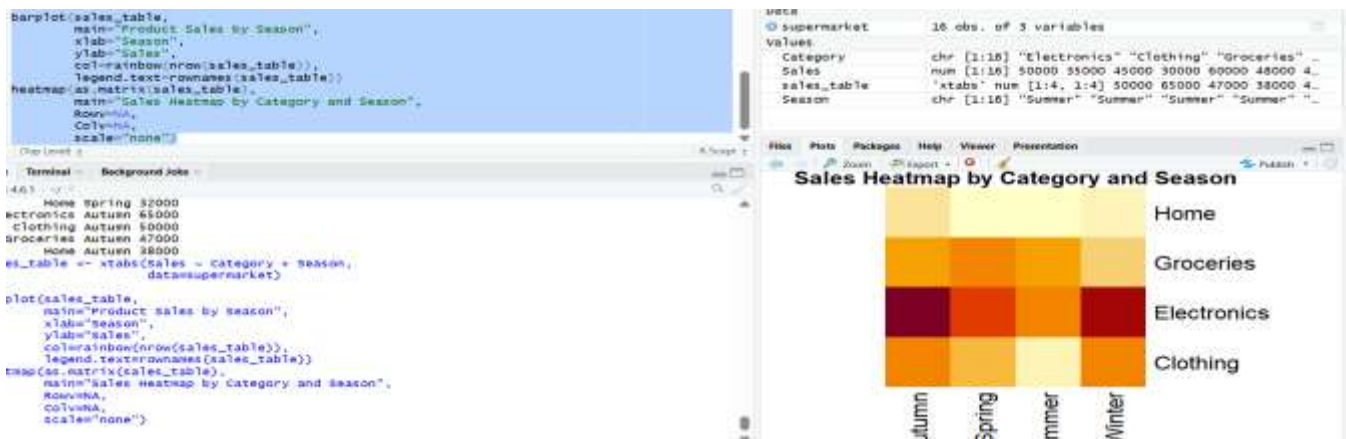


2. Heatmap

A heatmap can represent sales intensity for each combination of product category and season.

- Rows → Product Categories
- Columns → Seasons
- Color intensity → Sales amount

Use: Helps quickly identify high-sales and low-sales category-season combinations and makes seasonal trends easy to observe.

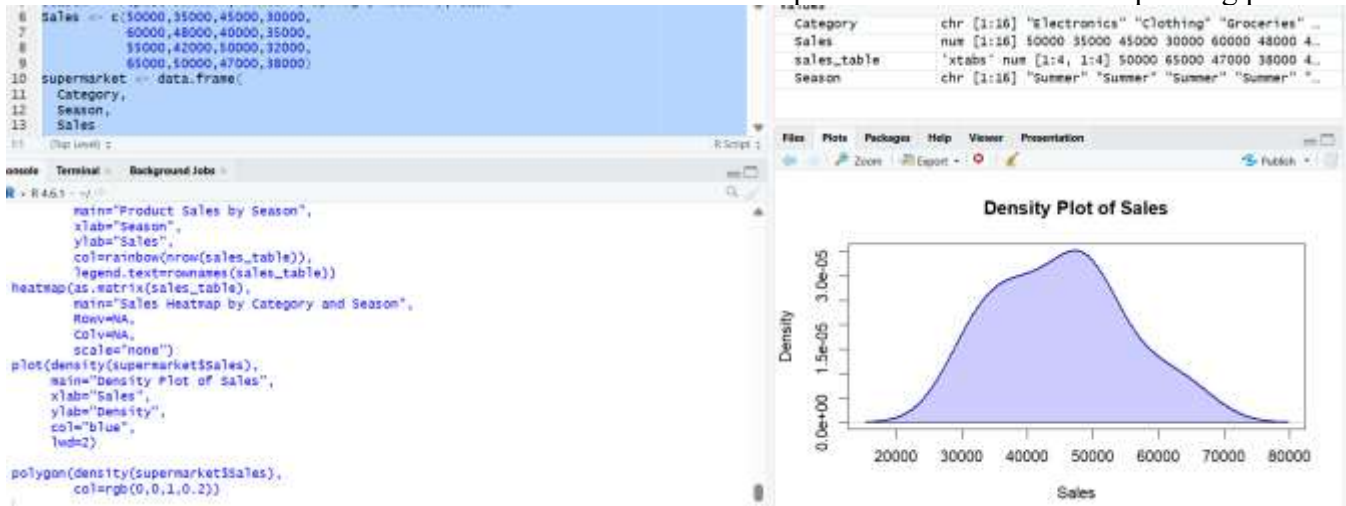


3. Density Plot

A density plot represents the overall distribution of customer spending or sales.

- X-axis → Sales
- Y-axis → Density

Use: Shows where most sales values are concentrated and helps understand the overall spending pattern.



4. Violin Plot

A violin plot compares sales distributions across different product categories.

- X-axis → Product Category
- Y-axis → Sales
- Shows distribution, spread, and concentration of sales.

Use: Helps compare sales variability among categories and identify categories with consistently high or low sales.



Conclusion

These visualizations help the supermarket identify customer preferences, seasonal purchasing trends, high-revenue categories, and variations in sales. The insights can support inventory planning, promotional strategies, pricing decisions, and identification of new business opportunities.

3.A smart city project gathers data on vehicle counts, traffic congestion levels, average speeds, and travel times across different locations and time periods. City planners aim to identify congestion hotspots and evaluate transportation efficiency.

Task: Propose visualization methods including geospatial maps, heatmaps, temporal plots, and multivariate visualizations. Describe how these visualizations can assist in traffic management, route optimization, and infrastructure planning.

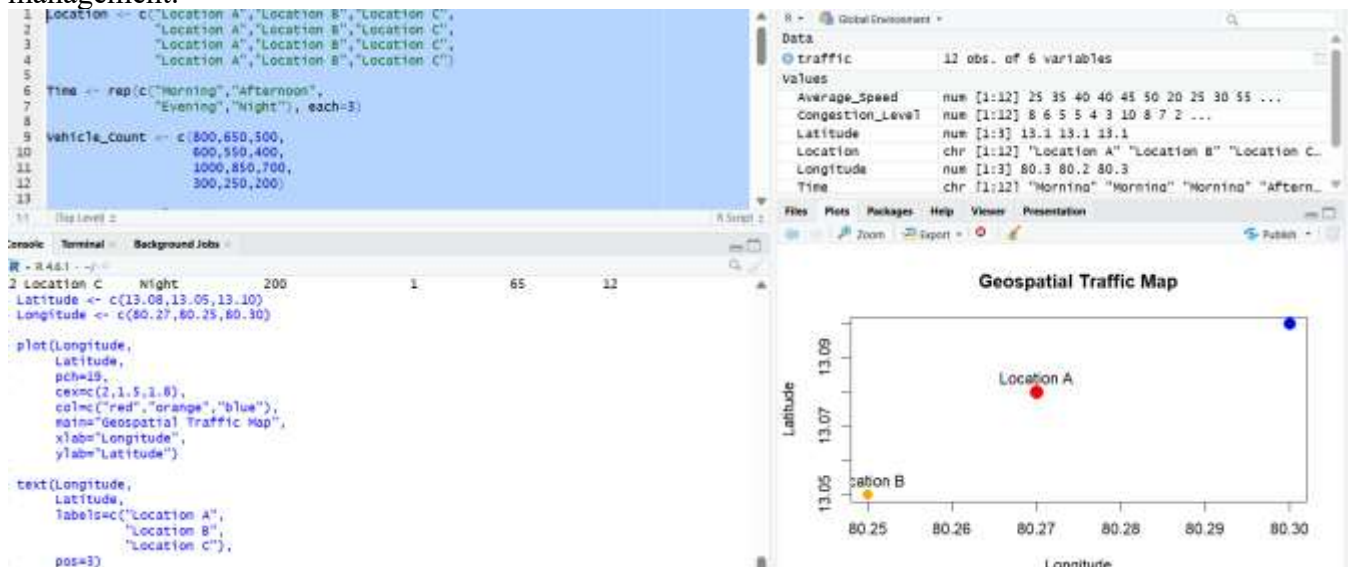
Aim: A smart city can use geospatial maps, heatmaps, temporal plots, and multivariate visualizations to analyse vehicle movement, congestion levels, average speeds, and travel times across different locations and time periods.

1. Geospatial Map

A geospatial map can display traffic conditions at different city locations.

- Location → Roads or city areas
- Point size → Vehicle Count
- Color → Congestion Level
- Different locations can be compared geographically.

Use: Helps city planners identify traffic congestion hotspots and areas requiring immediate traffic management.

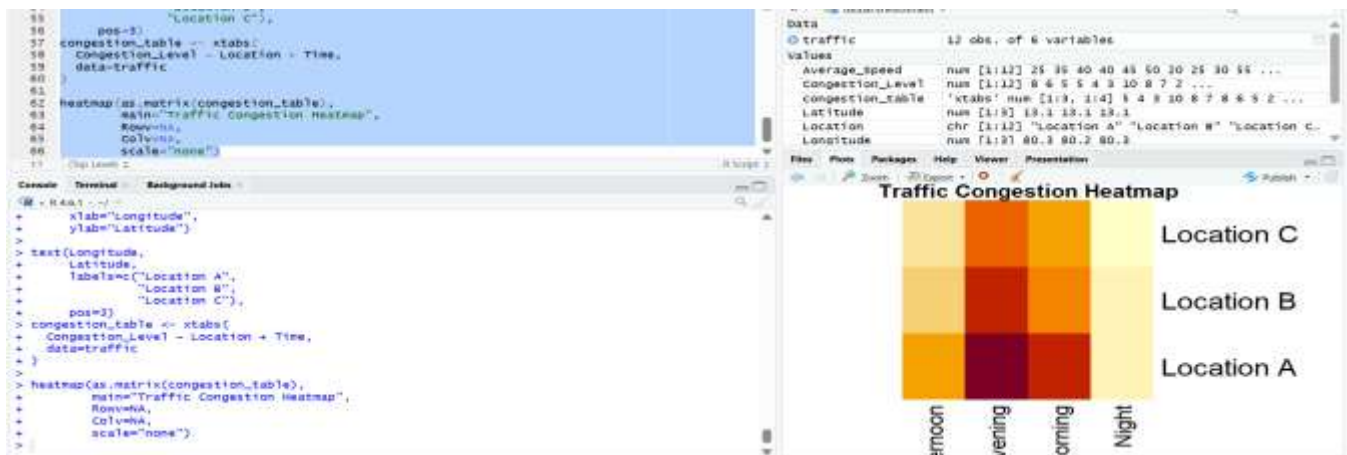


2. Heatmap

A heatmap can represent congestion levels across locations and different time periods.

- Rows → Locations
- Columns → Time Period
- Color intensity → Congestion Level

Use: Helps identify locations and time periods with high traffic congestion, such as morning and evening peak hours.

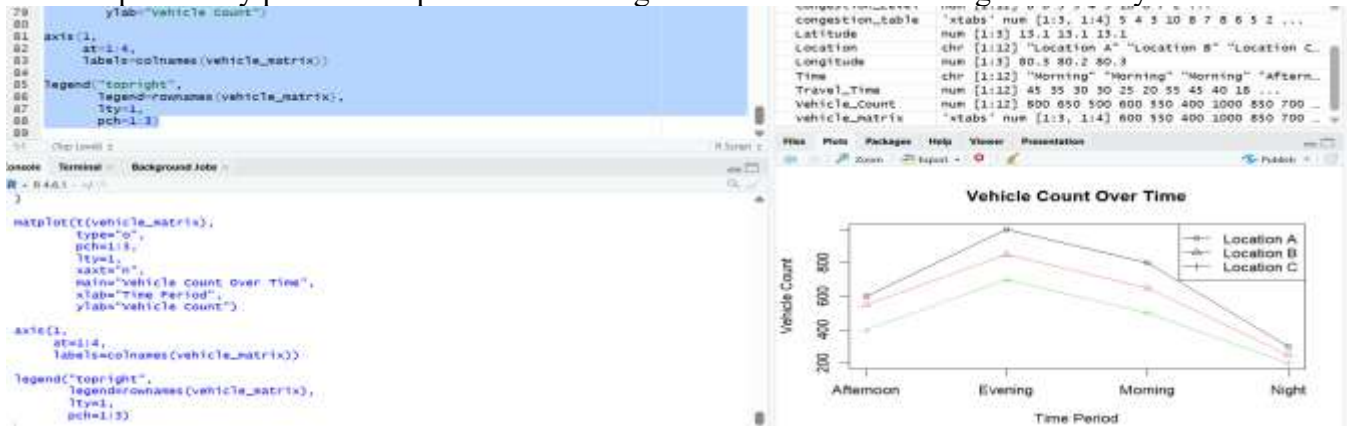


3. Temporal Plot

A line/temporal plot can show how traffic conditions change over time.

- X-axis → Time Period
- Y-axis → Vehicle Count
- Lines → Different Locations

Use: Helps identify peak traffic periods and changes in traffic volume throughout the day.

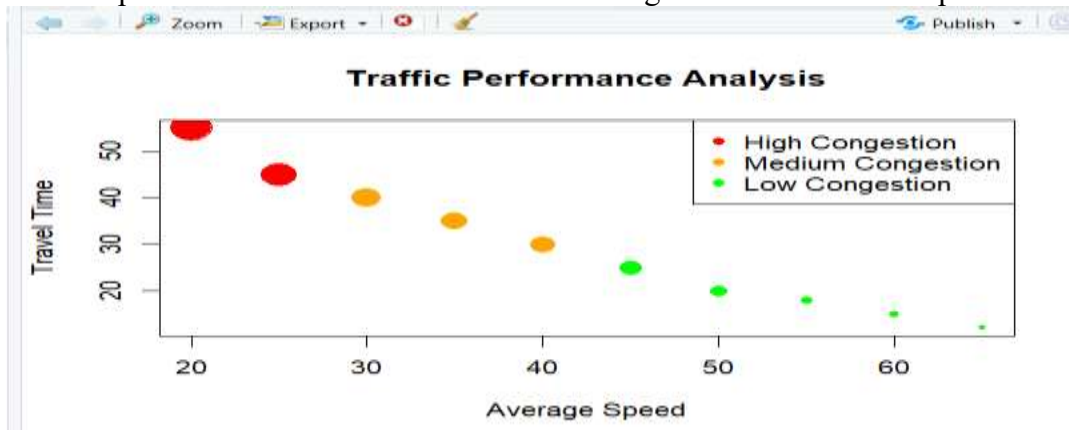


4. Multivariate Visualization

A bubble/scatter plot can compare several traffic variables simultaneously.

- X-axis → Average Speed
- Y-axis → Travel Time
- Bubble size → Vehicle Count
- Color → Congestion Level

Use: Helps determine how traffic volume and congestion affect vehicle speed and travel time.



Conclusion

These visualizations help city planners identify congestion hotspots, detect peak traffic periods, optimize routes, improve signal timing, and plan road infrastructure. They can also support decisions about road expansion, alternative routes, and public transportation.

4.A digital marketing company tracks user engagement metrics such as likes, shares, comments, viewing duration, and follower growth across multiple campaigns. The goal is to analyze relationships between engagement variables and campaign success.

Task: Suggest suitable visualization techniques such as scatter plots, heatmaps, network visualizations, and multivariate plots. Explain how these visualizations help identify influential factors, audience behaviour patterns, and effective marketing strategies.

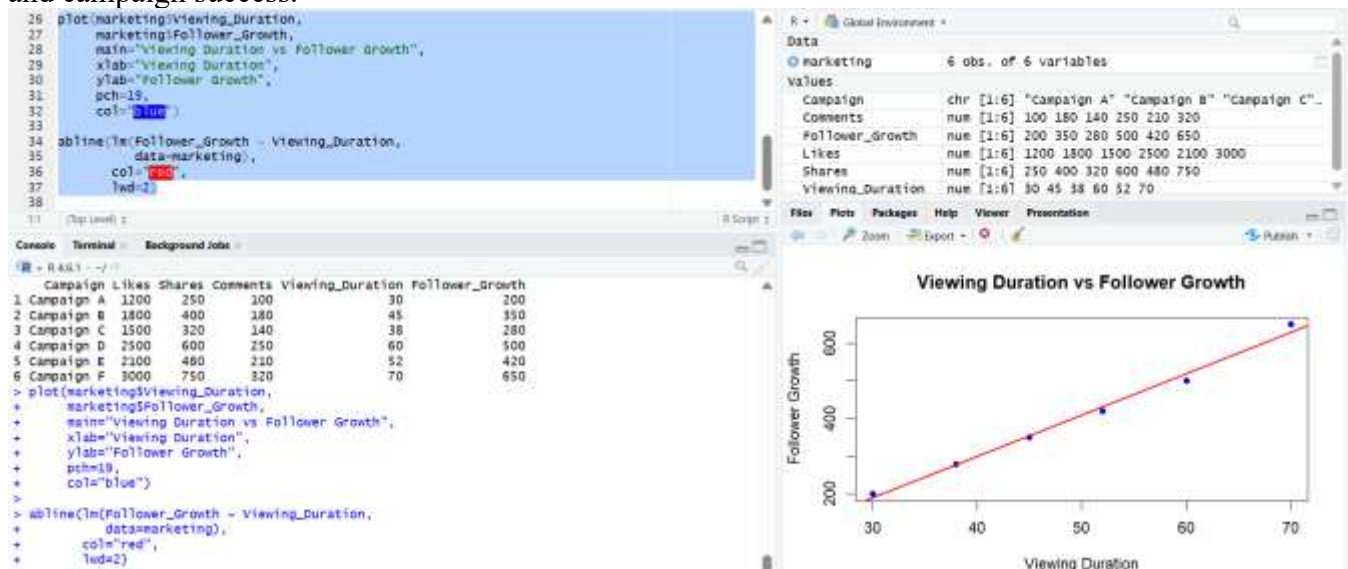
Aim: A digital marketing company can use scatter plots, heatmaps, network visualizations, and multivariate plots to analyse relationships between likes, shares, comments, viewing duration, follower growth, and overall campaign success.

1. Scatter Plot

A scatter plot can show the relationship between two engagement variables.

- X-axis → Viewing Duration
- Y-axis → Follower Growth
- Each point → One campaign

Use: Helps determine whether campaigns with longer viewing duration result in greater follower growth and campaign success.

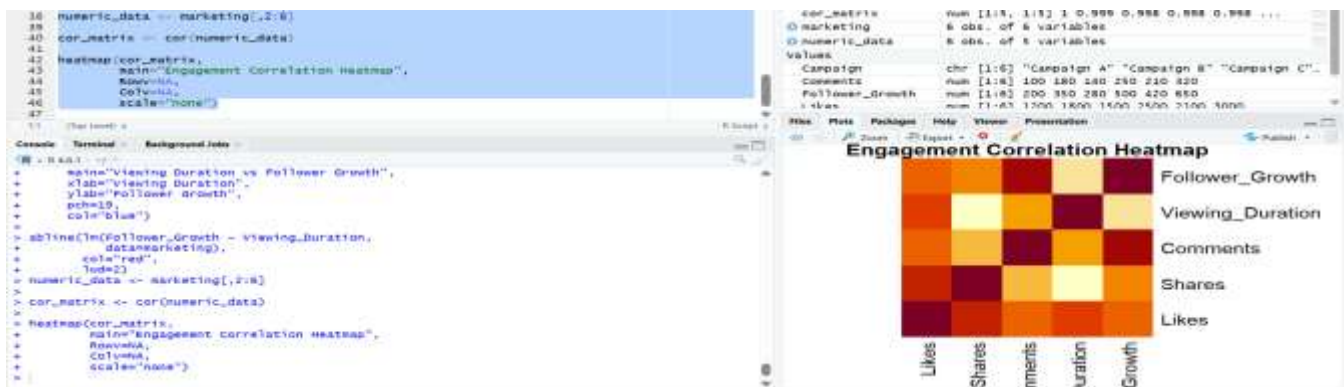


2. Heatmap

A correlation heatmap can display relationships among multiple engagement metrics.

- Variables → Likes, Shares, Comments, Viewing Duration, Follower Growth
- Color intensity → Strength of correlation

Use: Helps identify which engagement variables are strongly related and which factors have the greatest influence on campaign performance.



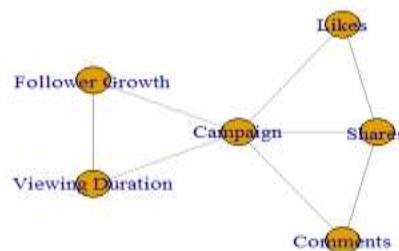
3. Network Visualization

A network visualization can represent relationships between campaigns and engagement activities.

- Nodes → Campaigns and engagement metrics
- Connections → Relationships between them

Use: Helps identify highly connected engagement factors and understand how different audience interactions contribute to campaign success.

Marketing Engagement Network

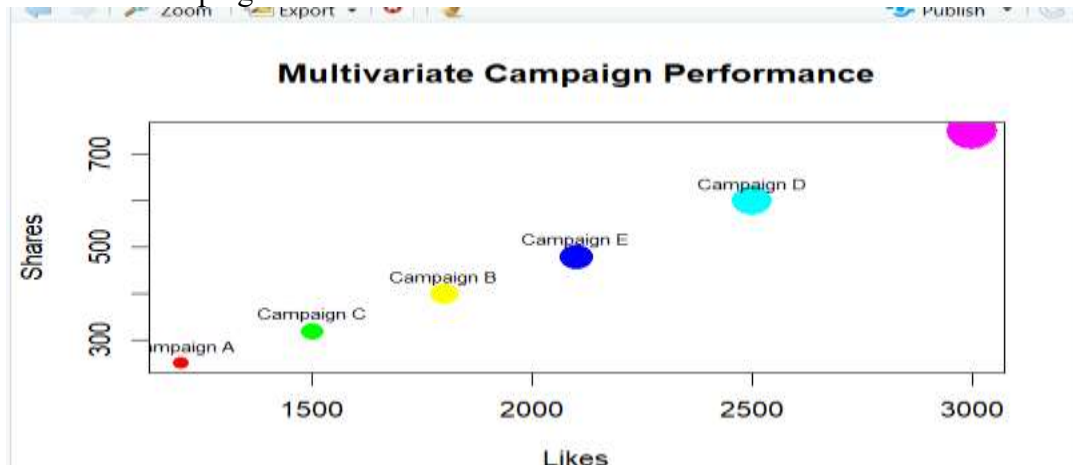


4. Multivariate Plot

A bubble plot can represent several engagement variables simultaneously.

- X-axis → Likes
- Y-axis → Shares
- Bubble size → Comments
- Color → Campaign success or follower growth

Use: Helps compare campaigns and identify combinations of engagement metrics associated with successful campaigns.



Conclusion

These visualizations help the company identify influential engagement factors, audience behavior patterns, successful campaigns, and relationships among engagement metrics. The findings can support better content creation, audience targeting, campaign optimization, and marketing strategies.

5.A healthcare organization maintains records of patients' ages, disease categories, treatment durations, recovery rates, and hospital visits. Medical administrators want to compare recovery distributions among patient groups and identify healthcare trends. Task: Select appropriate visualizations such as ridgeline plots, density plots, violin plots, histograms, and boxplots. Explain how these visualizations support disease analysis, treatment evaluation, and healthcare resource planning.

Aim: A healthcare organization can use **ridgeline plots, density plots, violin plots, histograms, and boxplots** to analyse patient age, disease categories, treatment duration, recovery rates, and hospital visits.

1. Ridgeline Plot

A **ridgeline plot** can compare recovery-rate distributions across different disease categories.

- X-axis → Recovery Rate
- Y-axis → Disease Category
- Each curve → Distribution for one disease group

Use: Helps compare recovery patterns across diseases and identify patient groups with better or poorer recovery outcomes.

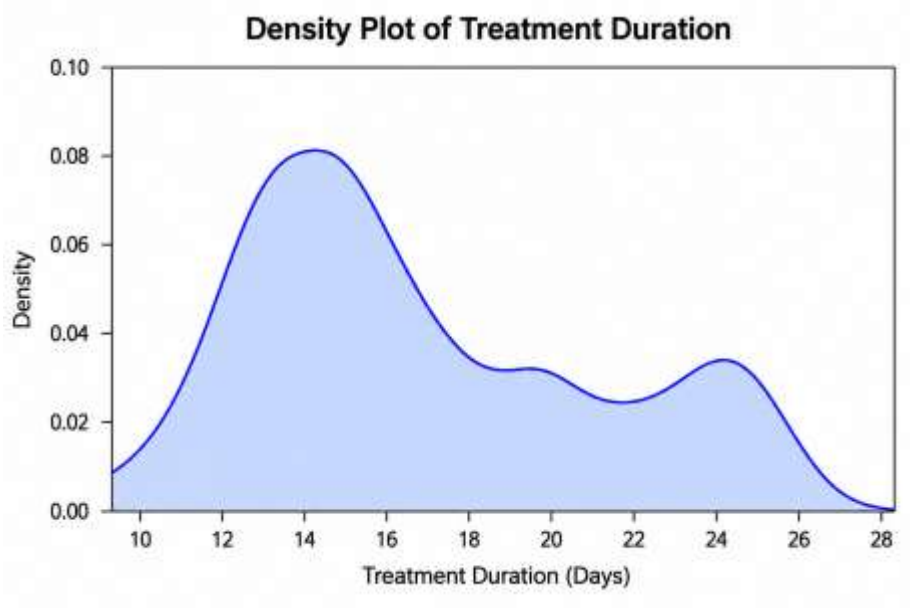


2. Density Plot

A **density plot** can show the distribution of treatment duration.

- X-axis → Treatment Duration
- Y-axis → Density

Use: Helps determine the most common treatment durations and identify whether some patients require unusually long treatment.

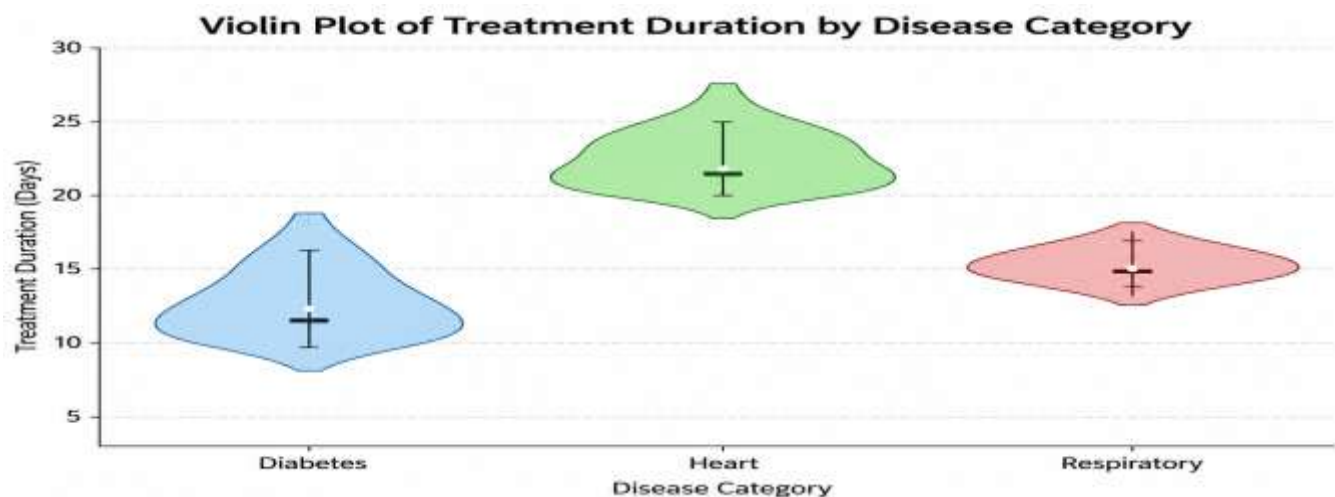


3. Violin Plot

A **violin plot** can compare treatment duration across disease categories.

- X-axis → Disease Category
- Y-axis → Treatment Duration
- Width → Concentration of patients

Use: Helps compare treatment duration, variability, and concentration among different disease groups.

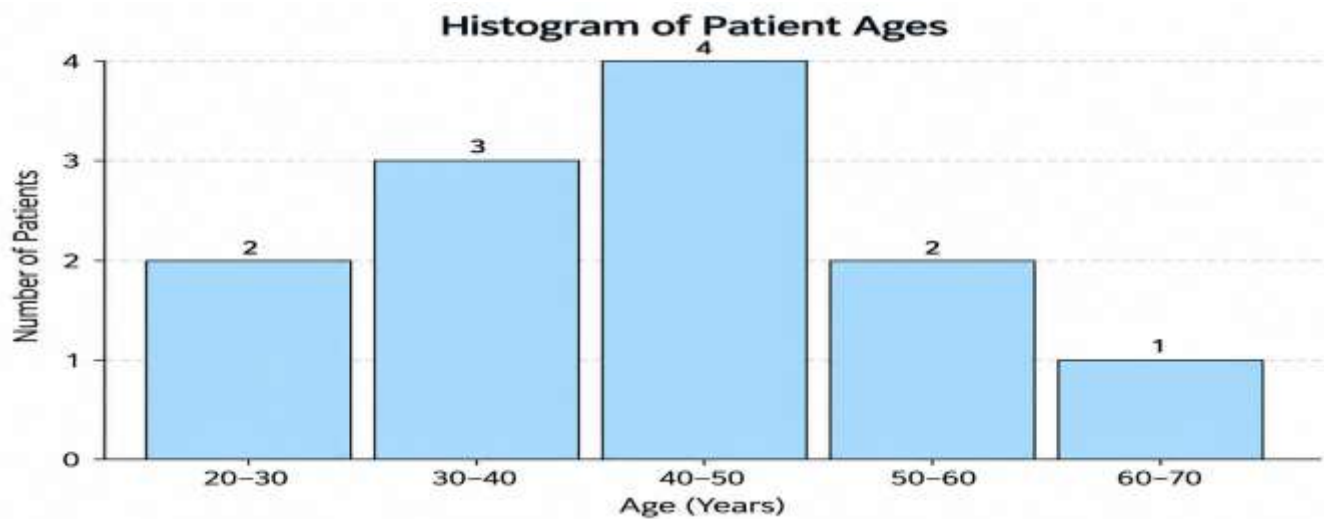


4. Histogram

A **histogram** can represent the distribution of patient ages.

- X-axis → Age groups
- Y-axis → Number of Patients

Use: Helps identify the most common patient age groups and supports age-specific healthcare planning.



Conclusion

These visualizations help healthcare administrators **compare disease patterns, evaluate treatment effectiveness, identify variations in patient recovery, and understand patient demographics**. The findings can support better allocation of **hospital beds, medical staff, treatment facilities, and other healthcare resources**.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context		

Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts		
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution		
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools		
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____